Approximating the score function using Optimal Transport

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We propose a novel framework for approximating the gradient flow of the relative entropy functional (with respect to the standard Gaussian) in Wasserstein space using Kantorovich potentials. By reformulating the Jordan-Kinderlehrer-Otto (JKO, [1]) scheme as a sequence of convex sub-problems, our approach leverages stochastic gradient descent (SGD) to efficiently solve each minimisation problem. We study the semi-discrete case, in which the source measure is seen as a discrete measure over the data points. In this scenario, we establish the convergence of the SGD iterates and provide error estimates. After following the approximated gradient flow (of the relative entropy) for some time, we present several methods for "inverting" this flow to generate new data points. Numerical experiments illustrate the potential of our method in learning complex data distributions.

Joint work with: Thomas Gallouët, Ziad Kobeissi.

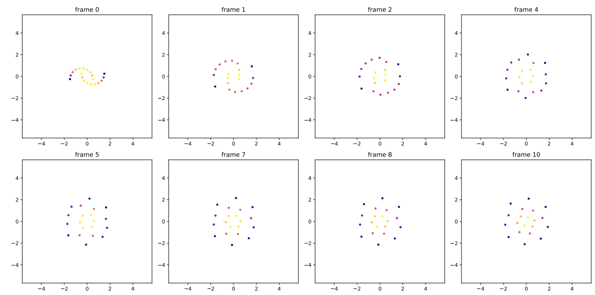

Figure 1: Forward: evolution of a point cloud under the gradient flow of the relative entropy, using our scheme.

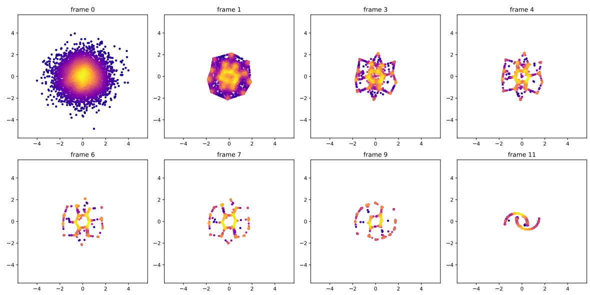

Figure 2: Backward: inverting the forward scheme to generate new samples.

References

[1] R. Jordan, D. Kinderlehrer and F. Otto. The Variational Formulation of the Fokker–Planck Equation. *SIAM Journal on Mathematical Analysis*, 10.1137/S0036141096303359, 1998

[2] F. Genans, A. Goddichon-Baggioni, F.-X. Vialard and O. Wintenberger. Stochastic Optimization in Semi-Discrete Optimal Transport: Convergence Analysis and Minimax Rate. *NeurIPS*, 10.48550/arXiv.2510.25287, 2025